

Network Infrastructure Day 1 Summary

Monday, October 5th, 2026

Overview

Key participants: QA Trainer Ninad Rampure leads the session and delivers demonstrations and CLI examples; active contributors include Arley Gibbons, Thomas Kent, Fabian, Liam Morris, Hamza, Rion Schofield, Simon Adams, Samad Alom, Ellis Critcheson, Melissa, and others.

Main topics and outcomes: The class continues the concept-first format with hands-on Packet Tracer labs and GitHub lab resources. Ninad covers device-level behaviour focusing on hubs versus switches, explaining how switches use MAC tables to micro-segment traffic and how ARP broadcasts operate within a broadcast domain. He introduces VLANs as the primary method to split a switch into multiple broadcast domains, explains inter-VLAN routing requires a layer-3 device, and shows voice VLAN and QoS considerations for VoIP.

Configuration and labs: Ninad demonstrates basic Cisco switch CLI commands (enable, config t), VLAN creation, assigning ports to access or trunk modes, and saving configurations, and confirms that deeper VLAN, trunking, and VTP labs will be performed in the next session. He reiterates expectations for Packet Tracer activities (build a simple LAN, propose a three-tier design) and reminds learners to extract GitHub zip files, check direct-message lab credentials, and use the recommended 'Skills for All' Packet Tracer login.

Commitments and next steps: Participants will complete the assigned Packet Tracer screenshots and the three-tier design deliverable; VLAN and VTP hands-on labs are scheduled for the next meeting as Ninad will demonstrate both GUI and CLI methods. The trainer will continue OSI-layer and STP configuration topics in upcoming sessions and share session notes for review.

Detailed Summary

Module structure and approach

Trainer Ninad explains the updated module format emphasizing concept understanding and active questioning rather than lecture-style slides. The course uses pre-questions for each concept so learners who have prior knowledge can progress faster and the class focuses on areas students need. Resources and lab materials are provided through a GitHub link and participants are instructed on how to download and use them.

- QA Trainer Ninad Rampure states the module now emphasizes concept understanding and uses pre-questions before each topic.
- QA Trainer Ninad Rampure explains the new format avoids long slide-driven lectures and targets gaps in learner knowledge.
- QA Trainer Ninad Rampure instructs participants to download lab files and the activity booklet from the provided GitHub link and to unzip files before editing.

- QA Trainer Ninad Rampure reminds students to check direct messages for lab usernames, passwords, and launcher details before the hands-on packet tracer activities.
- QA Trainer Ninad Rampure confirms labs will increasingly use Cisco Packet Tracer and that many upcoming activities are Packet Tracer-based.

Network types: peer-to-peer vs client-server

The group compares peer-to-peer and client-server network models, discussing architecture, scalability, security, management, and typical use cases. Participants and the trainer conclude client-server suits growing enterprises due to centralized management, role-based access control, and easier backup and monitoring, while peer-to-peer fits small, isolated, or ad-hoc scenarios (e.g., home LAN, torrenting, some gaming).

- Fabian characterizes peer-to-peer as suitable for small, decentralized networks.
- Liam Morris notes large corporate networks favor client-server for shared resources and scalability.
- Kai Standley and Blake highlight that client-server offers stronger security through individual logins while peer-to-peer can lack tracking and access control.
- Krystian emphasizes role-based permissions as a core mechanism enabling security in client-server environments.
- QA Trainer Ninad Rampure explains authentication, authorization, and accounting (AAA) are enforceable in client-server but not effectively in peer-to-peer.

Advantages and disadvantages analysis

Participants analyze cost, scalability, security, performance, and management trade-offs for both network types. While peer-to-peer has lower initial cost and works for small setups, client-server is preferred for business continuity, fault tolerance, centralized backups, monitoring, and easier policy enforcement as organizations scale.

- Liam Morris recommends client-server for the scenario because it supports growth despite higher initial cost.
- Arley Gibbons argues client-server enables domain features, device enrollment, and centralized security management.
- QA Trainer Ninad Rampure lists business continuity, fault tolerance, high availability, and centralized backups as client-server benefits.
- Participants cite peer-to-peer challenges including equipment variability, bottlenecks when many devices update from the internet, and difficulty enforcing traffic segmentation.

Real-world peer-to-peer examples

The group identifies practical examples where peer-to-peer is used, such as torrenting, some online gaming modes, and localized file sharing or update distribution. These examples illustrate how distributed resource sharing can improve availability and speed but also introduce legal and security concerns.

- Hamza identifies torrenting as a classic peer-to-peer example and explains seeding increases availability and download speed.
- Hamza also points out some online gaming uses peer-to-peer for confined game sessions.
- Liam Morris references Windows Server Update Services (WSUS) as a technique that centralizes updates to reduce external bandwidth usage, illustrating a client-server optimization versus pure peer-to-peer behavior.

Activity: recommending network model

Participants receive an activity to recommend peer-to-peer or client-server for a company of 50 employees doubling in two years. They must analyze scalability, cost, security, performance, and management and produce a justified recommendation supported by research. Most participants conclude client-server is the appropriate model for the described growth scenario.

- QA Trainer Ninad Rampure assigns the activity: evaluate peer-to-peer vs client-server for scalability, cost, security, performance, and management.
- Liam Morris and several others submit recommendations favoring client-server due to scalability and security despite higher upfront cost.
- QA Trainer Ninad Rampure instructs participants to save their work and join a later brainstorming session to compare justifications.

Resources and lab access details

Trainer provides practical instructions for accessing course materials, lab environments, and login details via direct messages and the GitHub repository. He warns about read-only behavior inside zip files and offers transcription of the session notes for later review.

- QA Trainer Ninad Rampure tells participants to click the GitHub repo, use 'download zip', and extract files to avoid read-only Word documents.
- QA Trainer Ninad Rampure shares that lab usernames, passwords, and launcher details are posted in direct messages for those who need them.
- QA Trainer Ninad Rampure confirms he transcribes and will share session notes so students can catch up on missed content.
- QA Trainer Ninad Rampure asks learners to confirm they can access the MyPC/Cisco launcher and warns about copying passwords without extra spaces.

Network zones: DMZ, extranet, intranet

The trainer teaches zone concepts (perimeter/DMZ, extranet, intranet), explaining placement, purpose, and how they protect internal resources. The DMZ hosts internet-facing services (e.g., web, VPN, authentication proxies) while extranet and intranet control external partner and internal employee access respectively, often separated by firewalls and VLAN/subnet routing.

- QA Trainer Ninad Rampure defines the perimeter/DMZ as the gateway zone for internet-facing services that should not have direct access to the internal network.

- Simon Adams and Blake identify firewalls and packet filtering as key protections between internal and external networks.
- QA Trainer Ninad Rampure explains extranet supports authenticated external users (customers/partners) via controlled portals and VPNs, while intranet is for internal employees.
- QA Trainer Ninad Rampure highlights use cases for DMZ-hosted services such as VPN servers, web servers, and remediation servers for noncompliant devices.

Security controls for extranet

The group discusses mechanisms to secure extranet access, stressing authentication, authorization, accounting, multi-factor authentication, and firewall rules. Conditional access and role-based policies are described as techniques to limit access by location, device compliance, or identity.

- Fabian suggests implementing authentication and authorization to protect extranet access.
- QA Trainer Ninad Rampure emphasizes the IAAA model (identification, authentication, authorization, accounting) and multifactor authentication for external access control.
- QA Trainer Ninad Rampure notes conditional access (e.g., location or device signals) and single sign-on as features that can improve secure productivity for trusted internal users.

Network topologies: physical vs logical

Trainer covers physical topologies (bus, star, ring, mesh) and their effects on performance and reliability. Participants identify collision domains, single points of failure (bus), central-node dependency (star), redundancy benefits (mesh), and ring behavior, and the trainer emphasizes that modern networks are predominantly star-based physically and logically.

- Samad Alom describes bus topology as a single backbone with a single point of failure and collision risk.
- Samad Alom and Arley Gibbons explain mesh topology provides multiple paths and higher fault tolerance.
- QA Trainer Ninad Rampure notes star topology centralizes traffic through a hub/switch, reducing collisions and forming the basis of most modern networks.
- QA Trainer Ninad Rampure introduces collision and broadcast domains as important considerations when choosing topologies and mentions half-duplex versus full-duplex behavior.
- QA Trainer Ninad Rampure explains ring uses token passing, enforces unidirectional forwarding, and becomes impractical at scale due to hops and latency.
- QA Trainer Ninad Rampure highlights mesh as resilient for wireless and IoT because nodes can reroute around failures and provide wider coverage and portability.
- QA Trainer Ninad Rampure states star is generally the most cost-effective physical design while mesh gives the best redundancy but at higher cost.

Network implementation types

Trainer reviews implementation scales—PAN, LAN, WLAN, CAN/campus, MAN/metropolitan, WAN—and maps typical technologies, distance limits, and carrier roles. He explains how LANs use Ethernet, MANs typically rely on fiber to interconnect city areas, and WANs rely on global infrastructure like submarine cables and telco providers.

- QA Trainer Ninad Rampure defines LAN as a geographically confined network using Ethernet and copper cabling for short distances.
- Thomas Kent and QA Trainer Ninad Rampure describe campus networks as multi-building LANs with area routers and a head router coordinating routing within the campus.
- QA Trainer Ninad Rampure explains MAN is citywide connectivity often provided by fiber (FTTC, FTTP) and WAN refers to global internet connectivity using submarine cables and telecom providers.
- QA Trainer Ninad Rampure highlights PAN (Bluetooth peripherals) and SAN (storage area networks/cloud storage) as other common network types and clarifies their use cases.
- QA Trainer Ninad Rampure notes WAN management is carried out by telecom and private providers and is only controllable up to the organization's edge router.

Zone-mapping activity: stakeholder scenarios

Learners complete an activity mapping five stakeholder scenarios (employee systems, partner collaboration, customer-facing website, online payment gateway, file transfer for vendors) to appropriate network zones with security justifications. The class volunteers and defends choices—internal systems to intranet, collaboration and partner file transfers to extranet, public websites and payment gateways to DMZ/perimeter—with recommended controls like MFA, HTTPS/TLS, firewalls, and isolation for sensitive payment systems.

- QA Trainer Ninad Rampure assigns five scenarios and asks learners to recommend zones and justify security/access decisions.
- Rion Schofield recommends intranet for employee systems and explains external access should be blocked.
- Hamza recommends extranet for partner collaboration and suggests MFA and user profiling for authentication and authorization.
- Ellis Critcheson places the customer-facing website in the perimeter/DMZ and highlights using firewalls and subnetting to protect internal resources.
- Fabian places the payment gateway in the perimeter/DMZ and emphasizes TLS/HTTPS and isolation from other public services to protect financial data.
- Melissa recommends extranet for partner/vendor file transfers and stresses restricted access plus MFA and authorization controls.

Lab readiness: Cisco Packet Tracer

Trainer confirms lab access and guides students to log into the remote lab launcher and Cisco Packet Tracer (preferring 'Skills for All' green link). He demonstrates how Packet Tracer emulates

network devices, cables, and endpoints and prepares learners for the upcoming activity to design a simple LAN that meets previous activity requirements.

- QA Trainer Ninad Rampure instructs learners to check direct messages for their assigned MyPC login, password, and access code and to avoid copying extra spaces when pasting passwords.
- QA Trainer Ninad Rampure advises using the 'Skills for All' (green) login for Cisco Packet Tracer to avoid authentication issues with Network Academy.
- QA Trainer Ninad Rampure explains Packet Tracer is a network design/emulation tool that models Cisco hardware, ports, cables, and endpoints for lab activities.
- QA Trainer Ninad Rampure asks learners to confirm when they reach the QA desktop and to launch Packet Tracer so he can demo building a simple LAN for the next activity.

Packet Tracer demo: building a LAN

Trainer gives a step-by-step Packet Tracer walkthrough highlighting device selection, port states, and endpoint configuration. He shows the device palette, selecting routers, switches, and endpoints, explains how to view port counts and link states, and outlines planning for a simple LAN lab exercise learners will replicate and screenshot for their workbook.

- QA Trainer Ninad Rampure shows the device palette in Packet Tracer and points out routers, switches, hubs, endpoints, and cable types on the bottom-left toolbar.
- QA Trainer Ninad Rampure demonstrates selecting a 24-port switch, hovering to view port counts and link state indications, and opening device config to inspect ports.
- QA Trainer Ninad Rampure explains endpoints (PCs and laptops) default to Ethernet in Packet Tracer and that wireless requires additional configuration.
- QA Trainer Ninad Rampure assigns students to build a basic LAN in Packet Tracer that supports the requirements identified in activity one and two and to save screenshots into their Word document.
- QA Trainer Ninad Rampure demonstrates manual versus automatic cable selection and explains crossover cable usage for connecting similar devices.
- QA Trainer Ninad Rampure shows how to power off devices to change hardware modules and how to add wireless adapters to laptops in Packet Tracer.

Cabling, IPs, and connectivity testing

Trainer covers common cable types, IP addressing basics, and methods to verify connectivity in Packet Tracer. He explains when to use console, straight-through, crossover, serial and fiber cables, how to set static or DHCP IPs on endpoints and servers, and how to verify connectivity using ping, Simple PDU, and Stimulation mode for packet-level analysis.

- QA Trainer Ninad Rampure lists console, straight-through (copper), crossover, serial, and fiber as common cable types and explains crossover is for connecting similar devices.
- QA Trainer Ninad Rampure instructs students how to manually connect cables to specific ports or use the automatic 'thunderbolt' tool to speed connections.

- Thomas Kent and QA Trainer Ninad Rampure confirm endpoints need IP addresses and Ninad explains servers and routers should use static IPs while endpoints may use DHCP.
- QA Trainer Ninad Rampure demonstrates ping via command prompt, Simple PDU for quick checks, and Stimulation mode to observe packet traversal and headers.
- QA Trainer Ninad Rampure highlights link color indicators (green active, orange configuring, red disabled) as a troubleshooting cue.

Wireless integration and DHCP

Trainer shows how to add wireless devices and configure a home-router access point for DHCP in Packet Tracer. He demonstrates configuring SSID and Wi-Fi password via the router GUI, connecting the access point to the switch, and confirming DHCP assignment on client devices, while noting servers should remain static.

- QA Trainer Ninad Rampure configures a home-router GUI in Packet Tracer to enable DHCP, set SSID and Wi-Fi password, and save settings.
- QA Trainer Ninad Rampure connects the access point to the switch and then shows how to join a laptop to the SSID and confirm DHCP assignment.
- QA Trainer Ninad Rampure advises that servers and network infrastructure devices should use static IPs while laptops and desktops can use DHCP.
- Fabian raises a DHCP failure issue and QA Trainer Ninad Rampure explains DHCP can be provided by a home-router or a server's services tab in Packet Tracer.

Redundancy and spanning tree

Trainer explains redundancy design choices and when Spanning Tree Protocol (STP) is required. He shows how adding multiple paths and uplinks increases fault tolerance and clarifies that switching loops at layer two require STP, while layer three routing reduces some loop concerns.

- QA Trainer Ninad Rampure demonstrates adding a second switch and alternate paths to remove single points of failure and increase redundancy.
- Liam Morris asks about loops from multiple uplinks and QA Trainer Ninad Rampure explains layer two switching loops require STP while layer three routing limits loop propagation.
- QA Trainer Ninad Rampure notes most switches have STP enabled by default and that STP configuration will be taught in a later module.
- QA Trainer Ninad Rampure points out blocked ports (orange) as an STP reaction to prevent loops and advises when to implement STP in designs.

Three-tier network hierarchy

Trainer introduces the three-tier architecture (core, distribution, access) as an industry best practice for scalable, resilient enterprise networks. He explains roles of each tier, full- and partial-mesh design choices for redundancy and bandwidth aggregation, and how the model supports performance, security zoning, and simplified growth.

- QA Trainer Ninad Rampure defines core as high-speed redundant backbone devices used primarily for inter-network traffic and internet egress.
- QA Trainer Ninad Rampure explains distribution switches aggregate access-layer traffic, perform routing between subnets, and enable policy enforcement.
- QA Trainer Ninad Rampure describes access layer as the end-user layer (layer two) connecting endpoints like PCs and printers.
- Hamza highlights resilience as a benefit and QA Trainer Ninad Rampure explains multi-path and redundant devices remove single points of failure.
- QA Trainer Ninad Rampure explains core and distribution are often full-mesh while distribution-to-access is partial mesh to avoid broadcast and switching issues.

Design activity: three-tier proposal

Learners are asked to propose a three-tier network in Packet Tracer that satisfies prior activity requirements (zones, client-server needs, scalability, security, performance, management). The task focuses on diagramming the design—routers, core and distribution layer-3 switches, access switches, and firewalls—with screenshots to justify choices rather than full device configuration.

- QA Trainer Ninad Rampure assigns a design task to create a three-tier network in Packet Tracer covering core, distribution, access, and zone placement for DMZ/extranet/intranet.
- QA Trainer Ninad Rampure advises including two core routers, redundant layer three switches, multiple distribution switches, access switches, and at least one firewall in the design.
- QA Trainer Ninad Rampure clarifies students should produce a diagram screenshot and justify how their design meets scalability, reliability, security, and management requirements.
- Rion Schofield asks whether activity two requires designs per scenario and QA Trainer Ninad Rampure responds no, produce a consolidated design addressing the key outcomes.

Network components introduction

Trainer begins an overview of common network devices and their functions to help students choose appropriate components during design. He frames the conversation as preparation for deeper device-level discussion and notes OSI layer coverage will be continued in the next session.

- QA Trainer Ninad Rampure introduces the upcoming discussion on networking components and explains selecting correct devices depends on understanding their capabilities.
- QA Trainer Ninad Rampure states OSI layer details will be covered in full on day two to provide context for device functions.

- QA Trainer Ninad Rampure begins by asking the class about hubs to prompt discussion on legacy and modern device usage.

Hubs vs switches

Trainer and participants compare hubs and switches to illustrate why switches replaced hubs in modern LANs. They explain hubs broadcast to all ports and operate half-duplex causing collisions and increased latency, while switches learn MAC addresses to forward traffic only to the destination port and provide separate collision domains per port, improving performance and scalability.

- Arley Gibbons explains a hub forwards incoming packets to all machines, letting endpoints determine whether the packet is for them.
- QA Trainer Ninad Rampure describes hubs as 'dumb' devices that do not maintain forwarding tables and therefore broadcast packets to every port.
- QA Trainer Ninad Rampure states hubs operate half-duplex so only one device can send or receive at a time, which creates collisions and latency.
- QA Trainer Ninad Rampure explains collisions trigger a jamming signal and require retransmission scheduling, increasing delays.
- QA Trainer Ninad Rampure notes a hub represents a single collision domain even if multiple hubs are interconnected, extending the collision domain.
- Arley Gibbons and QA Trainer Ninad Rampure suggest hubs can still be useful for one-to-many tasks such as bulk firmware updates or multicast-like replication.
- QA Trainer Ninad Rampure compares switches by explaining they learn MAC addresses and forward traffic only to the correct port, reducing unnecessary broadcasts.
- Thomas Kent highlights that switches reduce unnecessary traffic by sending frames only to the destination port using MAC address tables.
- QA Trainer Ninad Rampure explains each switch port is its own collision domain, so switches eliminate the shared-collision problem hubs cause.

Switch behavior and VLANs

Trainer expands on switch operation, ARP behavior, broadcast vs collision domains, and the role of VLANs for logical segmentation and QoS. He demonstrates how switches build MAC tables, how ARP broadcasts resolve MAC addresses within a broadcast domain, and how VLANs and layer-3 devices segment traffic to improve security, performance, and manageability.

- QA Trainer Ninad Rampure explains switches perform micro-segmentation and forward frames by destination MAC address rather than broadcasting to all ports.
- QA Trainer Ninad Rampure states ARP uses a broadcast MAC (FF:FF:FF:FF:FF:FF) to discover a host's MAC address and that broadcast traffic stays within a single switch broadcast domain.
- QA Trainer Ninad Rampure notes one switch typically equals one broadcast domain unless VLANs or routers are used to segment traffic.

- QA Trainer Ninad Rampure describes VLANs as virtual segments that partition a single physical switch into multiple broadcast domains and isolate traffic between departments.
- QA Trainer Ninad Rampure explains inter-VLAN communication requires a layer-3 device (router or layer-3 switch) to route between VLANs.
- QA Trainer Ninad Rampure highlights VLANs enable QoS and voice/data separation by using voice VLANs and sub-interfaces for prioritizing VoIP traffic.
- QA Trainer Ninad Rampure explains switches evolve with features like PoE, managed vs unmanaged operation, per-port configuration (duplex, shutdown), and VTP for VLAN propagation.

Switch configuration demo

Trainer demonstrates basic CLI steps to configure a Cisco switch and create VLANs, showing both GUI and CLI approaches. He covers entering enable/configure modes, creating VLANs, assigning ports to access or trunk modes, and saving configuration, and schedules hands-on VLAN and VTP labs for the next session.

- QA Trainer Ninad Rampure demonstrates entering enable and global configuration mode (config t) as the first steps to configure a Cisco switch.
- QA Trainer Ninad Rampure shows VLAN creation via the 'vlan ' command and naming the VLAN, and then assigning ports to VLANs in interface configuration.
- QA Trainer Ninad Rampure explains access ports carry untagged traffic for endpoints while trunk ports carry multiple VLANs between switches.
- QA Trainer Ninad Rampure notes most ports default to access mode and that trunking is used between switches to transport VLAN traffic.
- QA Trainer Ninad Rampure points out saving the running configuration is required to persist settings and that he will teach VLAN/trunk CLI lab exercises tomorrow.

Session wrap-up and next steps

Trainer closes the session by summarizing remaining topics and scheduling VLAN/VTP labs for the next meeting. He thanks participants for engagement, confirms continuation tomorrow, and stops the session after checking for final questions.

- QA Trainer Ninad Rampure says VLAN and VTP configuration labs will be covered in the next session or the first thing tomorrow.
- QA Trainer Ninad Rampure thanks participants for interaction and confirms they will resume with activity four and remaining labs tomorrow.
- QA Trainer Ninad Rampure asks if there are any final questions and ends the session after no further queries.
- Participants (Blake, Rion Schofield, Simon Adams, Thomas Kent, Fabian) thank the trainer as the session concludes.